r/artificial

The AI Boom Could Use a Shocking Amount of Electricity

Posted by u/NuseAI • • 0 points • 33 comments

- The rapid growth of artificial intelligence (AI) could lead to a significant increase in global electricity consumption, according to a peer-reviewed analysis published in Joule.

- The analysis estimates that if current trends continue, AI could drive the demand for electricity in data centers to consume at least 85.4 terawatt-hours annually, which is more than what many small countries use in a year.

- AI is energy-intensive, with both the training and inference phases requiring a significant amount of energy.

- The size of AI models, such as large language models, and the location of data centers also contribute to energy usage.

- Factors such as cooling requirements and the type of hardware used can impact energy consumption.

Source : https://www.scientificamerican.com/article/the-ai-boom-could-use-a-shocking-amount-of-electricity/

Comments

> **u/tibimon** • 1 points • 2023-10-16 06:21 UTC
>
> Your car hast a way higher energy consumption per Person. So everyone complaining get rid of your car then we can talk about shutting down ai.

> **u/SexyJimBelushi** • 1 points • 2023-10-16 03:15 UTC
>
> Maybe enough AI will tell enough rich people they could get more richer by going more energy efficient and it will work itself out.

> **u/Electronic_Crazy8122** • 1 points • 2023-10-14 16:52 UTC
>
> not once my work gets off the ground. once the military tech I'm working on gets past phase 2 (TRL8) I'll be developing a commercial version. 20% power reduction to start for a single DNN convolution layer and later it will be many layers deep for one DNN and the entire CNN will be rolled in. My tech will

render GPUs obsolete because it'll be way faster, beat quantum to the punch, and draw a tiny fraction of the power.

u/ViolentNun • 1 points • 2023-10-14 04:55 UTC

Damn, lamps in movies use real electricity :0

u/Superb_Raccoon • 1 points • 2023-10-14 01:16 UTC

This was written by an AI, right?

u/Zimmax • 1 points • 2023-10-14 00:40 UTC

\> The analysis estimates that if current trends continue, AI could drive the demand for electricity in data centers to consume at least 85.4 terawatt-hours annually

Is that even realistic? TPU production is not a simple process and can't be scaled up indefinitely. Is that taken into account?

u/GFrings • 1 points • 2023-10-13 23:21 UTC

It IS using a shocking amount of electricity

u/inteblio • 1 points • 2023-10-13 23:16 UTC

Energy Is Not A Sin

> **u/futro-sleep** • 1 points • 2023-10-16 22:53 UTC
>
> Who’s saying it is? Or is putting it in caps a way of saying something ironically?
>
>> **u/inteblio** • 1 points • 2023-10-17 00:25 UTC
>>
>> "Carbon emissions are the real numbers that we care about when it comes to environmental impact. Energy demand is one thing, but is it coming from renewables? Is it coming from fossil fuels?"

u/Aquaritek • 1 points • 2023-10-13 23:16 UTC

I heard somewhere that "The human body generates more bioelectricity than a 120-volt battery and over 25,000 BTUs of body heat.." and that combining that with a special form of fusion.. well.. the machines may have all the energy they could ever need.

u/jessicaHacks • 4 points • 2023-10-13 22:58 UTC

This complaint is raised about almost every technological innovation tbh.

We'd probably save an order of magnitude more energy just by not eating beef.

u/squareOfTwo • 2 points • 2023-10-13 21:58 UTC

energy is the lifeblood of civilization.

> **u/futro-sleep** • 1 points • 2023-10-16 22:52 UTC
>
> Sure, but also a liveable climate. It’s no accident that agriculture started at the same time the climate stabilized to close to where it is now, around 10,000 years ago. Destabilize it and see what happens. Furthermore, energy being a lifeblood, perhaps we should use it wisely and not squander what we don’t know for certain we can renew, before we can actually do that.